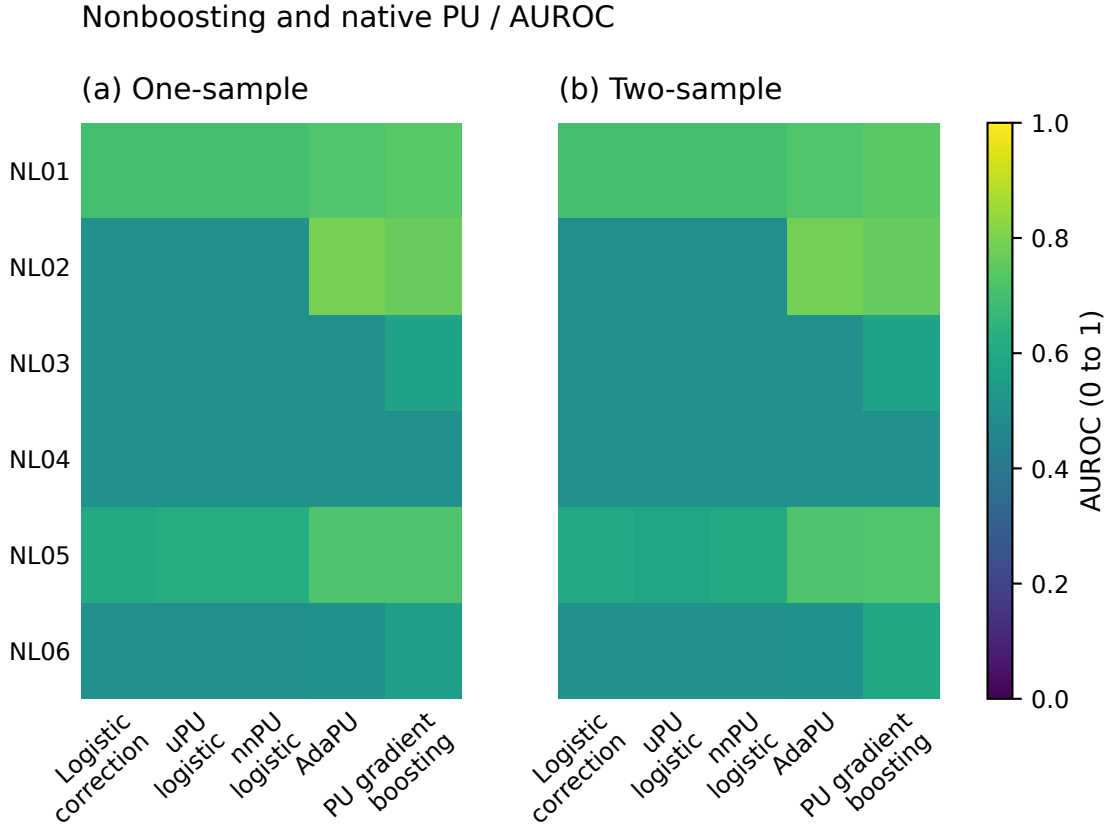
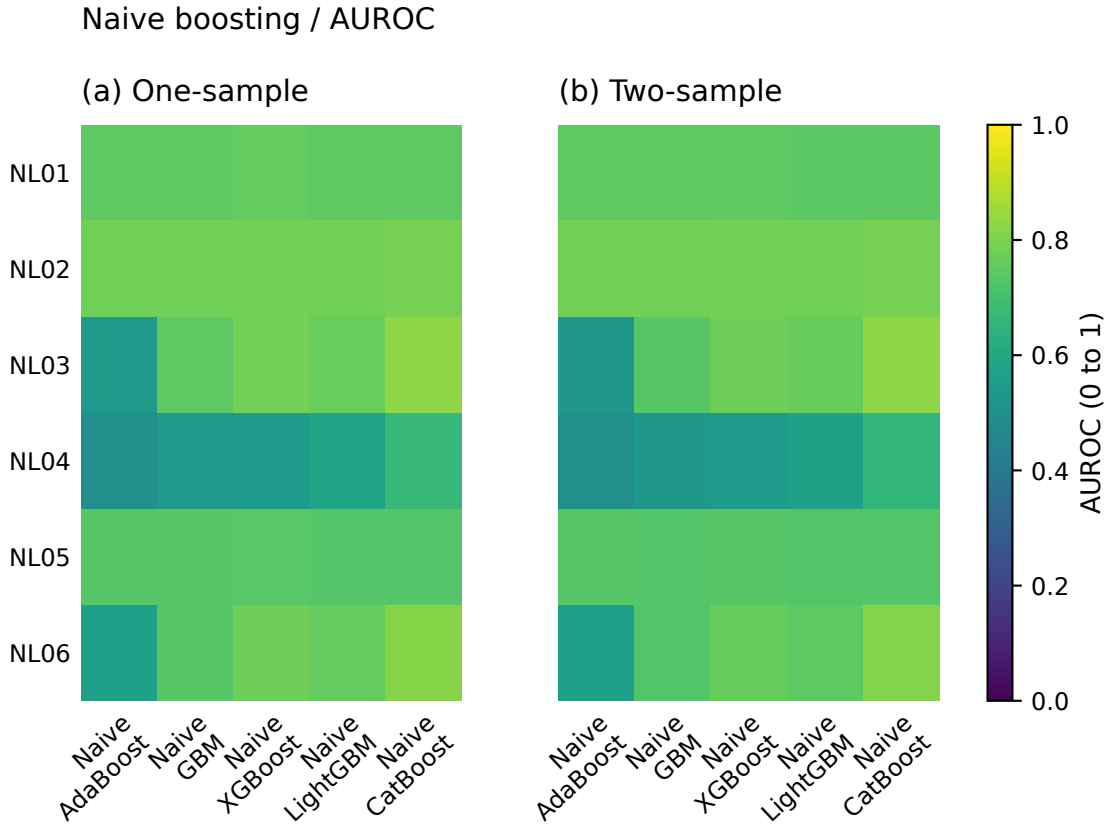
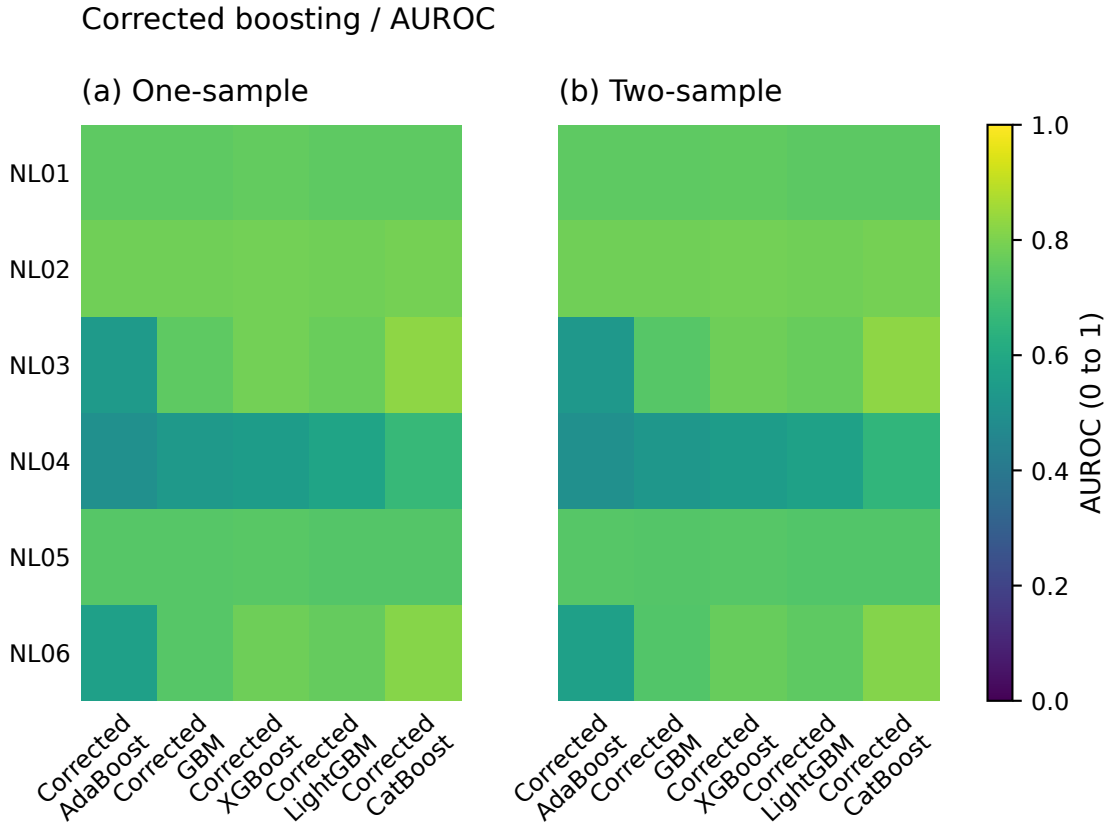
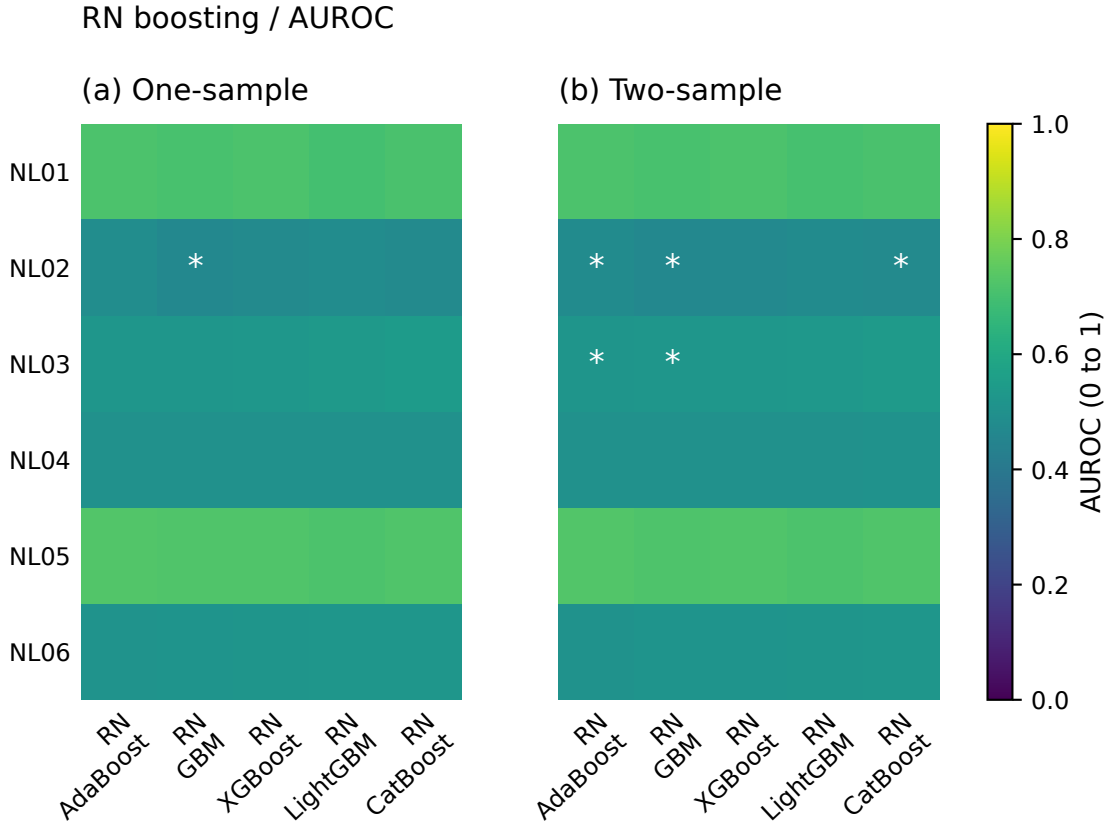
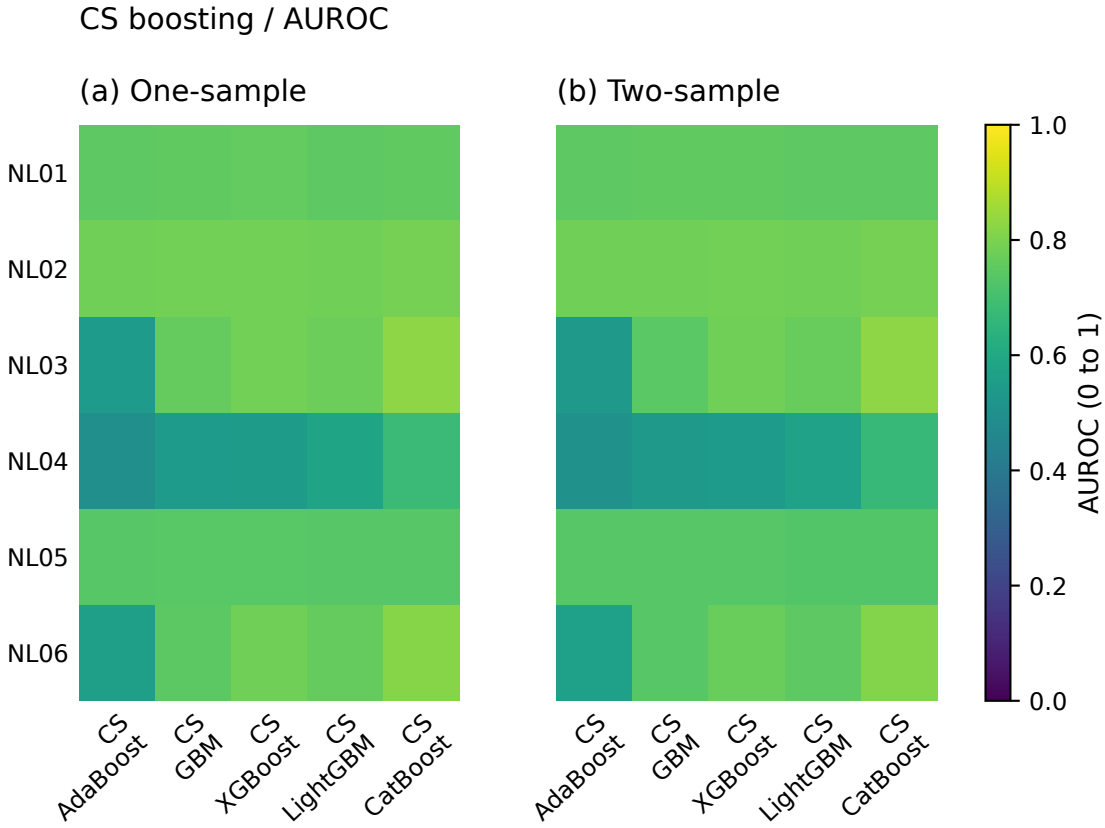
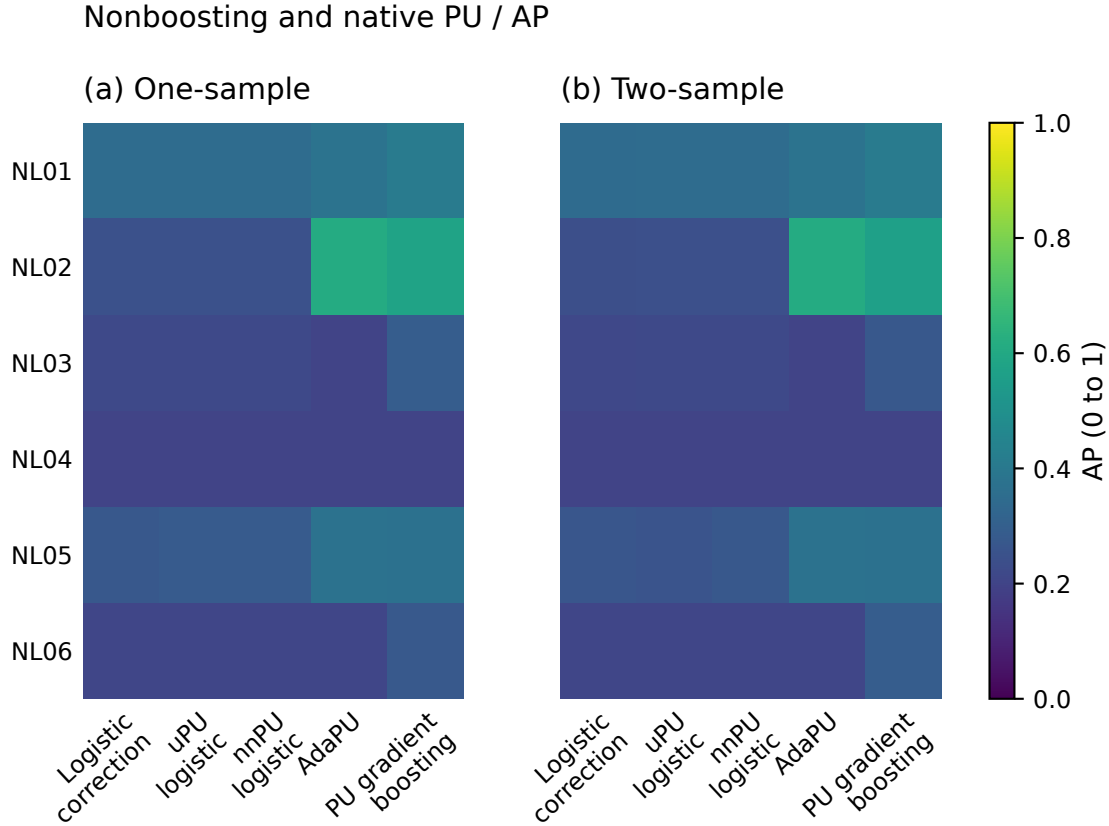

Nonboosting and native PU / AUROC
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

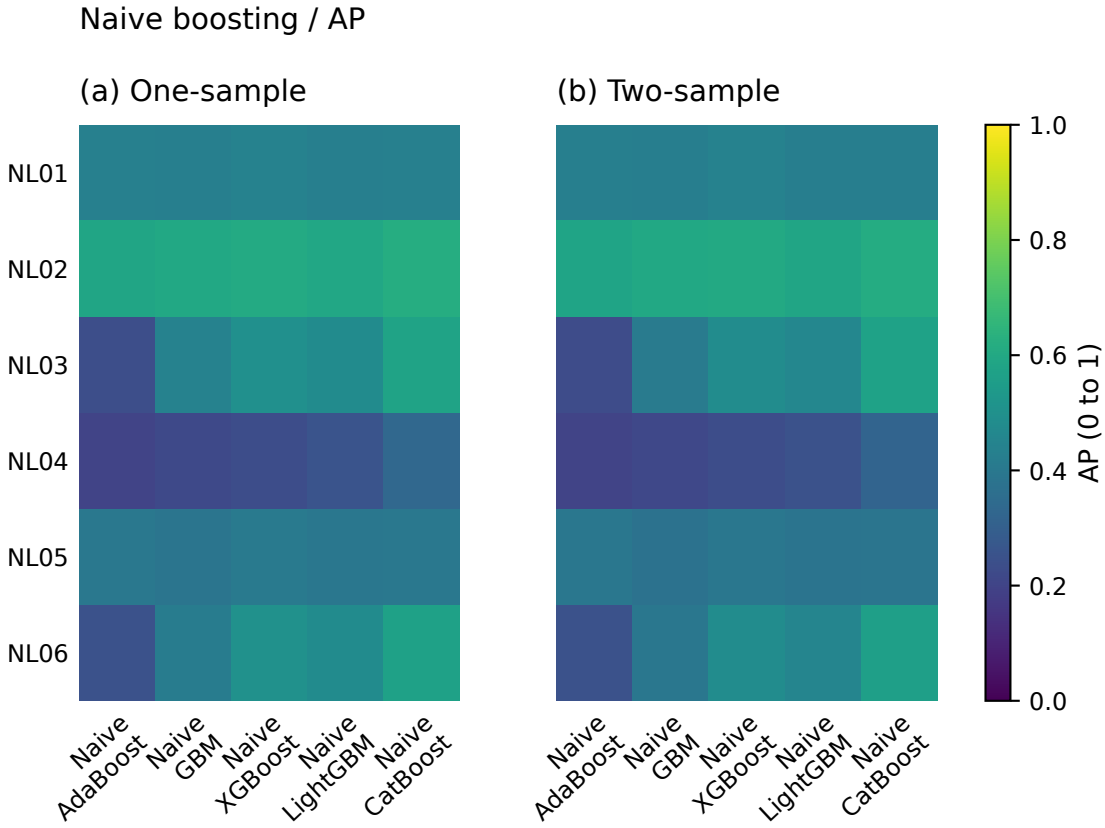

Naive boosting / AUROC
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

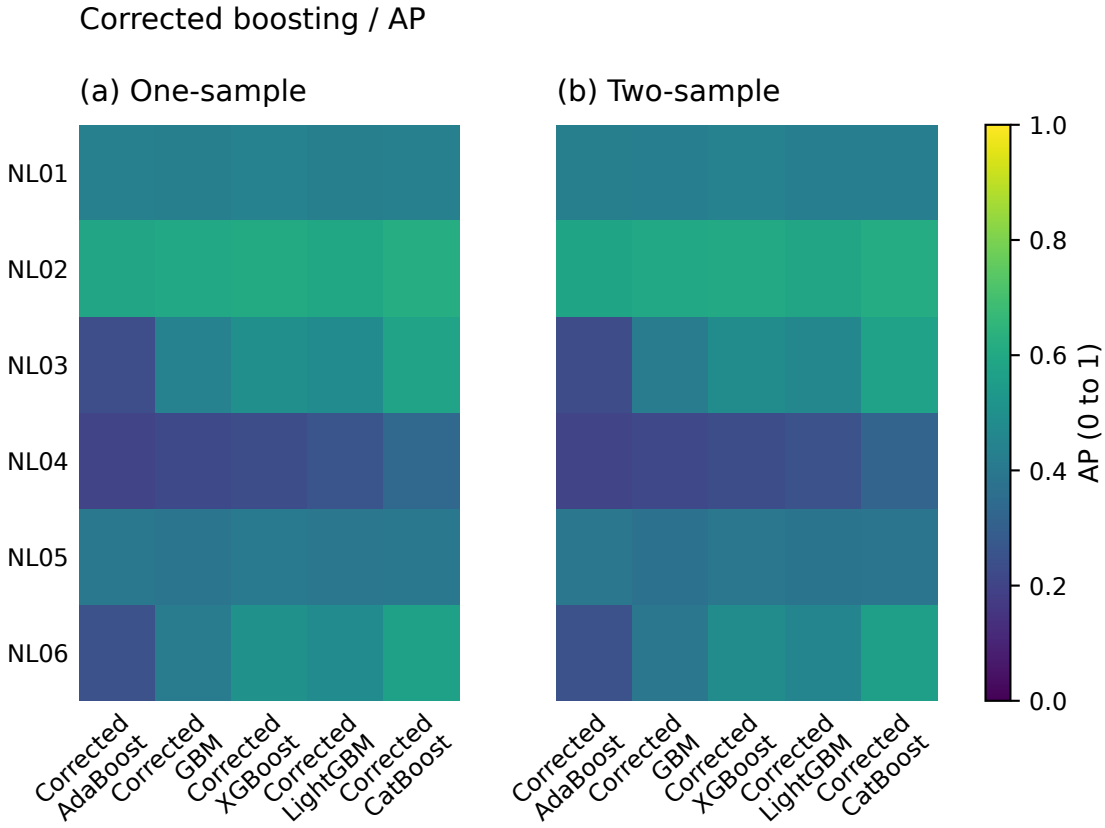

Corrected boosting / AUROC
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
Corrected AdaBoost
Corrected GBM
Corrected XGBoost
Corrected LightGBM
Corrected CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

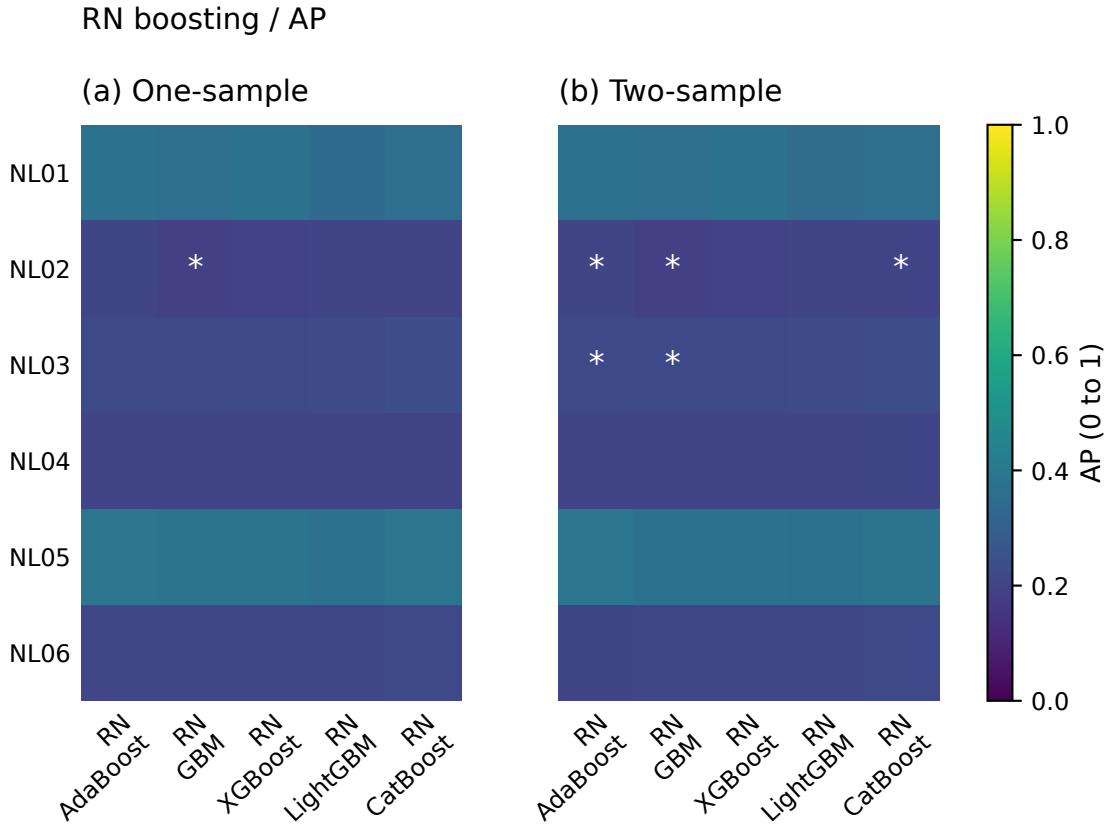

RN boosting / AUROC
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
RN AdaBoost
RN GBM
RN XGBoost
RN LightGBM
RN CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0

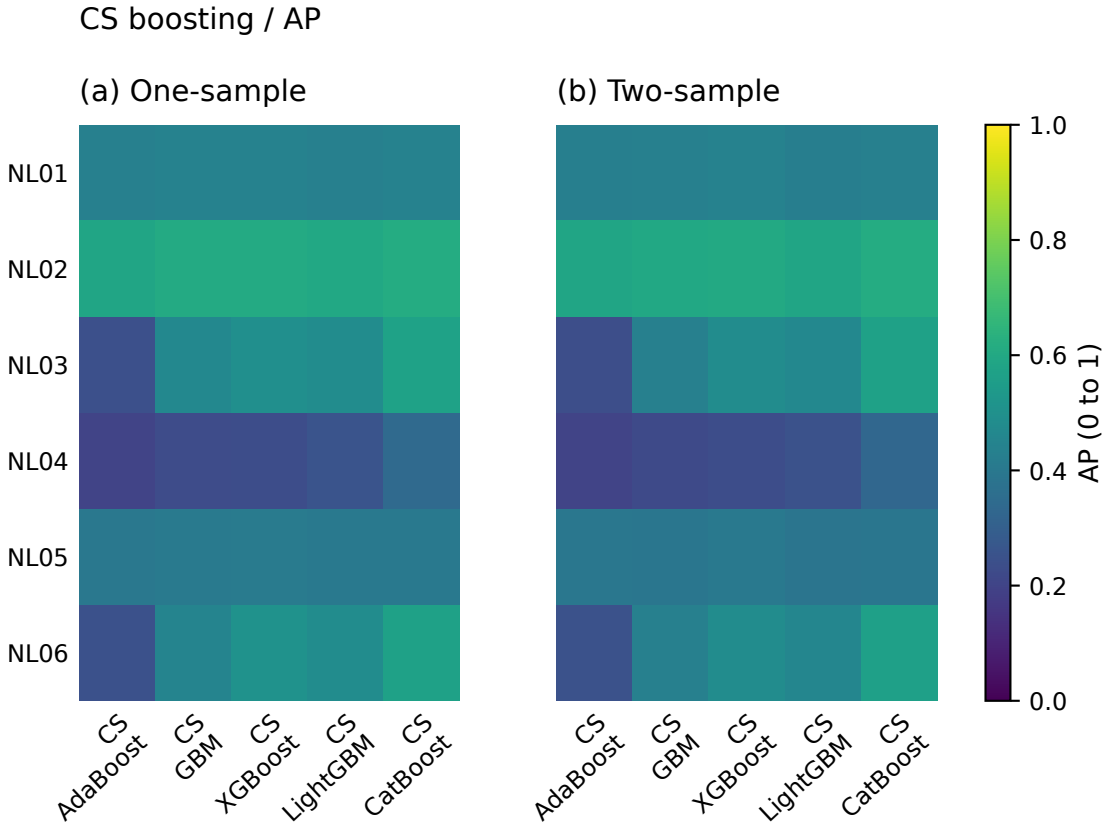

CS boosting / AUROC
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
CS AdaBoost
CS GBM
CS XGBoost
CS LightGBM
CS CatBoost
AUROC (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Nonboosting and native PU / AP
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
Logistic correction
uPU logistic
nnPU logistic
AdaPU
PU gradient boosting
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Naive boosting / AP
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
Naive AdaBoost
Naive GBM
Naive XGBoost
Naive LightGBM
Naive CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


Corrected boosting / AP
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
Corrected AdaBoost
Corrected GBM
Corrected XGBoost
Corrected LightGBM
Corrected CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


RN boosting / AP
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
RN AdaBoost
RN GBM
RN XGBoost
RN LightGBM
RN CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0


CS boosting / AP
(a) One-sample
(b) Two-sample
NL01
NL02
NL03
NL04
NL05
NL06
CS AdaBoost
CS GBM
CS XGBoost
CS LightGBM
CS CatBoost
AP (0 to 1)
1.0
0.8
0.6
0.4
0.2
0.0